

AP CALCULUS AB - UNIT 2

Guided Notes 2.4: Basic Derivative Rules



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we differentiate powers x^n for integer, rational, and permitted real exponents?

- Differentiate powers x^n for integer, rational, and permitted real exponents.
- Rewrite radicals and reciprocals as powers before differentiating.
- Evaluate derivatives and tangent slopes without unnecessary expansion.
- Apply constant, sum, difference, and constant-multiple rules efficiently.
- Differentiate polynomial and power combinations term by term.
- Interpret linearity of differentiation and avoid distributing rules that do not exist.
- Differentiate the four foundational transcendental functions in radians.
- Combine these derivatives using linearity and power rules.
- Use exact values, domains, and context when evaluating derivatives.
- Differentiate $\tan x$, $\cot x$, $\sec x$, and $\csc x$.
- Evaluate derivatives at exact angles and identify domain restrictions.
- Combine the remaining trigonometric derivative rules with prior linearity, Product, and Quotient Rules.

Key Knowledge, Vocabulary, and Notation

Power Rule: Power Rule $\frac{d}{dx}x^n = nx^{n-1}$ wherever the original function and derivative are defined.

Power Rule: Radical notation $\sqrt[q]{x^p} = x^{p/q}$ on the appropriate real domain.

Power Rule: Negative exponent $x^{-m} = 1/x^m$ for $x \neq 0$.

Power Rule: Domain check A derivative formula does not erase restrictions inherited from the original function.

Linearity Rules: Constant Rule $\frac{d}{dx}(C) = 0$.

Linearity Rules: Constant Multiple Rule $\frac{d}{dx}[cf(x)] = cf'(x)$.

Linearity Rules: Sum/Difference Rules $(f \pm g)' = f' \pm g'$.

Linearity Rules: Linearity $D[af + bg] = aDf + bDg$ for constants a, b .

Core Transcendental Derivatives: Sine and cosine $(\sin x)' = \cos x$ and $(\cos x)' = -\sin x$ when x is measured in radians.

Core Transcendental Derivatives: Natural exponential $(e^x)' = e^x$.

Core Transcendental Derivatives: Natural logarithm $(\ln x)' = 1/x$ for $x > 0$.

Core Transcendental Derivatives: Unit-circle discipline Use exact trigonometric values whenever the angle is standard and no calculator is allowed.

Core Transcendental Derivatives: Unit-circle discipline Use exact trigonometric values whenever the angle is standard and no calculator is allowed.

Tangent, Cotangent, Secant, and Cosecant: Tangent $(\tan x)' = \sec^2 x$.

Tangent, Cotangent, Secant, and Cosecant: Tangent $(\cot x)' = -\csc^2 x$.

Tangent, Cotangent, Secant, and Cosecant: Cotangent $(\sec x)' = \sec x \tan x$.

Tangent, Cotangent, Secant, and Cosecant: Secant $(\csc x)' = -\csc x \cot x$.

Tangent, Cotangent, Secant, and Cosecant: Cosecant

Concept Development and Strategy

1. **Power Rule.** Lower the exponent by one after multiplying by the original exponent. Keep exact fractional exponents unless the problem requests radical form.
2. **Power Rule.** Check where the function is real and defined. For example, $x^{1/2}$ is real for $x \geq 0$, while its derivative $1/(2\sqrt{x})$ requires $x > 0$.
3. **Power Rule.** A limit that matches the derivative definition of a power function can often be evaluated immediately by the Power Rule.
4. **Linearity Rules.** Differentiation is linear: constants can be pulled through the derivative, and sums can be differentiated term by term.
5. **Linearity Rules.** First simplify algebraically when it reduces work, especially when common factors or quotients of powers can be rewritten. Do not invent a rule such as $(fg)' = f'g'$; products require the Product Rule in Topic 2.8.
6. **Linearity Rules.** After differentiating, evaluate only if a particular input is requested. Keeping symbolic structure often reduces arithmetic errors.
7. **Core Transcendental Derivatives.** The standard trigonometric derivative formulas assume radians. Degree measure introduces an additional conversion factor.
8. **Core Transcendental Derivatives.** For sums and constant multiples, differentiate each term separately. Composite functions are postponed to the Chain Rule in Unit 3.
9. **Core Transcendental Derivatives.** Preserve the domain of $\ln x$. A derivative formula evaluated outside the original function domain is meaningless.
10. **Tangent, Cotangent, Secant, and Cosecant.** Keep signs attached to cotangent and cosecant derivative rules. A compact memory pattern is: tangent and secant are positive; cotangent and cosecant are negative.
11. **Tangent, Cotangent, Secant, and Cosecant.** Use unit-circle values to evaluate exactly. Check that the original function is defined at the requested angle before evaluating its derivative.
12. **Tangent, Cotangent, Secant, and Cosecant.** These formulas can be derived from $\tan = \sin/\cos$, $\cot = \cos/\sin$, $\sec = 1/\cos$, and $\csc = 1/\sin$ using earlier rules.

Worked Example: Power Rule - Polynomial power

Differentiate $f(x) = x^7$.

Answer and Reasoning

$$f'(x) = 7x^6.$$

Worked Example: Power Rule - Radical and reciprocal

Differentiate $g(x) = \sqrt[3]{x^5} + \frac{1}{x^4}$.

Answer and Reasoning

$$g(x) = x^{5/3} + x^{-4}, \text{ so } g'(x) = \frac{5}{3}x^{2/3} - 4x^{-5}.$$

Worked Example: Power Rule - Derivative-definition limit

Evaluate $\lim_{h \rightarrow 0} \frac{(3+h)^8 - 3^8}{h}$.

Answer and Reasoning

This is the derivative of x^8 at 3, so the value is $8 \cdot 3^7$.

Worked Example: Linearity Rules - Polynomial

Differentiate $f(x) = 7x^5 - 3x^2 + 9x - 4$.

Answer and Reasoning

$$f'(x) = 35x^4 - 6x + 9.$$

Worked Example: Linearity Rules - Fractional powers

Differentiate $g(x) = 4x^{3/2} - 6x^{-1/3} + 11$.

Answer and Reasoning

$$g'(x) = 6x^{1/2} + 2x^{-4/3}.$$

Worked Example: Linearity Rules - Linearity from known derivatives

If $f'(2) = 5$ and $g'(2) = -1$, find the derivative of $H(x) = 3f(x) - 4g(x) + 9$ at 2.

Answer and Reasoning

$$H'(2) = 3(5) - 4(-1) = 19.$$

Worked Example: Core Transcendental Derivatives - Mixed basic derivatives

Differentiate $f(x) = 4 \sin x - 3 \cos x + 2e^x - 5 \ln x$.

Answer and Reasoning

$$f'(x) = 4 \cos x + 3 \sin x + 2e^x - \frac{5}{x}.$$

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Power Rule: Differentiate x^9 .

2. Power Rule: Differentiate x^{-3} .

3. Power Rule: Differentiate $f(x) = x^{7/3} - x^{-2}$.

4. Power Rule: Find $f'(16)$ for $f(x) = x^{3/4}$.

5. Power Rule: Differentiate $f(x) = \frac{x^5+1}{x^3}$ after rewriting.

6. Power Rule: For $f(x) = x^p$ with $p \neq 0$, solve $f'(1) = f(1)$.

7. Power Rule: Using only the Power Rule, show that the derivative of $x^{m/n}$ is $\frac{m}{n}x^{m/n-1}$ for positive x by differentiating $y^n = x^m$ informally and solving for y' .

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. Power Rule: $\frac{d}{dx}x^{-5}$ equals

(A) $-5x^{-6}$

(B) $-5x^{-4}$

(C) x^{-6}

(D) $5x^4$

2. Power Rule: The derivative of $\sqrt[3]{x^2}$ for $x \neq 0$ is

(A) $\frac{2}{3}x^{-1/3}$

(B) $\frac{3}{2}x^{1/3}$

(C) $2x^{-2/3}$

(D) $\frac{1}{3}x^{-2/3}$

3. Power Rule: A calculator evaluation of $f'(7.2)$ for $f(x) = x^{1.7}$ is closest to

(A) 1.7

(B) 4.01

(C) 6.77

(D) 28.7

Common Errors and AP Precision

- State the derivative notation, input value, sign, and units when interpreting a rate in context.
- Check differentiability before using a tangent slope, and keep the order of Product and Quotient Rule terms accurate.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Tangent, Cotangent, Secant, and Cosecant: Prove $(\csc x)' = -\csc x \cot x$ from the reciprocal form $1/\sin x$.

Original instructional resource prepared for Education City High School. Not an official College Board publication.